FRESH WATER AND PUBLIC AUTHORITY IN MADAGASCAR: A LEGAL ANALYSIS OF A POORLY GUARANTEED VITAL RIGHT

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ABSTRACT

The 1999 Water Code (Law No. 98-029) makes fresh water part of the common heritage of the Malagasy nation, and international human rights law, to which the Constitution gives precedence over ordinary statutes, grounds a right of access to it. In practice, this right is proclaimed far more than it is delivered. This article confronts Madagascar's dense and formally modern water legislation with the reality of public action. Combining doctrinal and comparative legal analysis with operational data from the Atsimo-Andrefana region (twenty monitored supply sites, 268 infrastructure worksites and 165 multiple-use boreholes), it identifies a threefold failure: regulatory bodies created by decree but never made operational; a decentralization that gives municipalities contracting authority without the means to exercise it; and the de facto replacement of the public authority by donors and NGOs in service delivery. The contribution is twofold. Empirically, the article brings into legal analysis field data on rural water services in South-West Madagascar that legal scholarship has rarely used. Conceptually, it shows that the ineffectiveness of the right to water reflects a deficit of administration rather than of legislation, and it turns this diagnosis into an operational roadmap specifying actions, responsible actors, timelines and indicators. Guaranteeing this vital right calls less for new statutes than for a redesign of governance: an operational regulator, sustainable and climate-informed financing, controlled contracting, a stronger water police and a recognized place for customary water institutions.

KEYWORDS

Right to Water, Water Code, Decentralization, Water Governance, Legal Pluralism, Madagascar

1. INTRODUCTION

"It is pleasant to find water when one is thirsty" (*Tsara raha mangetaheta ka mahita rano*): legal scholarship has used this Malagasy proverb as an epigraph [1, p. 65], and it states the matter in negative terms, since the value of water shows itself when water is lacking. What was popular wisdom has become, through the Constitution and the Water Code, a matter of positive law. Yet in a country where a large part of the population still depends on unimproved sources, the proverb sounds less poetic than alarming. Fresh water is at once a vital resource, a scarce economic good and an instrument of power: whoever controls its capture, treatment and distribution exercises concrete authority over people's living conditions. Water thus raises a legal question and a question of public authority that cannot be dealt with separately.

This intertwining of law and power is no accident. Historically, the mastery of water has always been an attribute of power: from ancient hydraulic societies to modern States, controlling the resource means controlling economic and social life. In Madagascar, this dimension is compounded by strong symbolic and customary meanings, water being invested with ritual and communal significance that positive law cannot ignore. The shift from customary, localized management to State-based, codified management is in itself a displacement of power, and its tensions are visible today in the difficulties of implementing the legal framework.

The value of a renewed legal analysis lies precisely here. Malagasy water law is not a normative desert. As early as 1999 the country adopted a Water Code regarded as advanced, followed by a series of ambitious implementing decrees, and it has accepted international commitments that recognize a right of access to water. Yet the distance between the texts and lived reality is considerable. The present study takes this distance, between the formal recognition of a vital right and its actual guarantee, as its object.

1.1. Terminological Clarifications: Fresh Water, Public Authority, Vital Right

Three notions organize the argument. *Fresh water* refers here to non-saline water, whether surface water (rivers, lakes, reservoirs) or groundwater (aquifers), considered both as a natural resource and as a good serving human needs, first of all drinking, sanitation and hygiene, the field that donors group under the acronym WASH (Water, Sanitation and Hygiene) [2, chs. 1–2]. Marine and brackish waters, which follow a different legal regime, are excluded.

Public authority is understood in the administrative-law sense, as the set of bodies holding public power: the central State and its deconcentrated services; public establishments and specialized administrative authorities, foremost among them the bodies in charge of regulating and managing the resource; the decentralized territorial authorities (*Collectivités territoriales décentralisées*, CTDs); and, straddling the public and commercial spheres, the national utility JIRAMA. This acronym stands for *Jiro sy Rano Malagasy*, literally “Malagasy Electricity and Water” (*jiro*: light, electricity; *rano*: water). The analysis covers what these authorities do as well as what they fail to do: in administrative law, omissions, abstentions and organizational shortcomings are not legally neutral and may engage the liability of the public authority. A glossary of the Malagasy terms and acronyms used in the article is provided after the conclusion.

Finally, a *vital right* denotes a right whose deprivation directly threatens the life and dignity of the person. The right to water belongs to this category: it conditions the exercise of other fundamental rights such as the rights to life, to health and to adequate food, and it is at once individual (each person's right to access safe water in sufficient quantity) and collective (the nation's interest in preserving a common heritage). Its qualification as a fundamental right has precise legal consequences: it gives rise, on the part of the public authority, to obligations to respect, to protect and to fulfil [3, §§ 20–29].

1.2. Spatial and Temporal Framework: Madagascar Facing Water and Climate Constraints

The study considers Madagascar in its present situation. The island presents a well-known hydrological paradox: its water resources are abundant at the national scale, yet it experiences sharp spatial and seasonal disparities and a severe lack of infrastructure for mobilizing and distributing the resource. The South and South-West concentrate these tensions. The Atsimo-Andrefana region (literally “South-West”), on which the case study in Section 4 is based, illustrates them: semi-aridity, irregular rainfall, groundwater that is at times deep or mineralized, and a scattered rural settlement pattern that raises the unit cost of service.

Climate change adds to this structural constraint. Recent work on the South-West documents more intense droughts, a disrupted rainfall calendar and the growing vulnerability of the agro-pastoral systems that depend on it [4]. Growing scarcity shifts the legal debate: a question of management becomes a question of arbitration between competing uses (human consumption, livestock watering, irrigation), and water law takes on an increasingly redistributive function. Health risks follow from the same scarcity: poor water quality and intermittent service create exposure that, according to vulnerability studies, weighs first on the poorest households [5].

1.3. Significance of the Subject

The significance of the subject is threefold. On the legal plane, it confronts the theory of sources and of the hierarchy of norms with the test of effectiveness: what is a proclaimed fundamental right worth if the structures meant to implement it remain virtual? The question renews the classic reflection of administrative law on public service, the allocation of competences and the liability of the public authority for its failings. On the social plane, access to water directly shapes public health, school attendance (particularly of girls) and the condition of women, who traditionally bear the chore of fetching water. On the economic plane, water management involves pricing, cost recovery and long-term investment, which belong to development economics and to the strategic management of natural resources: the financial sustainability of the service conditions its durability, and hence the very effectiveness of the right.

1.4. Research Question, Contribution and Outline

The research question follows: how does Malagasy public law treat the water resource, and why does the present institutional architecture fail to make this fundamental right effective? Our hypothesis is that the ineffectiveness of the right to water in Madagascar results less from gaps in the law than from the way public authority is organized and resourced. In short, the deficit is one of administration rather than of legislation.

The article seeks to add to existing work in three ways. Malagasy water law has so far been studied mainly through its texts and their convergence with other legal orders [1], or through sector reviews written by practitioners [2]. First, the article confronts this normative framework with operational data from a semi-arid region, a type of material that legal scholarship has rarely used. Second, it reads these data with the tools of administrative law (public service, allocation of competences, liability of the public authority) combined with those of development economics (unfunded mandates, information asymmetry, cost recovery). Third, it draws on comparative experience and on the interface between statutory and customary norms to formulate recommendations that specify actions, responsible actors, timelines and indicators.

Section 2 sets out the methodology and data. Section 3 shows that the legal recognition of the right to water, ambitious as it is, comes up against the limits of Madagascar's institutional framework. Section 4 examines the challenges of public action, from the paralysis of decentralization to an operational roadmap for redesigning water governance. Section 5 discusses the limitations of the study, and Section 6 concludes.

2. METHODOLOGY AND DATA

2.1. Legal Method and Corpus

The study follows a doctrinal approach: it identifies the applicable sources, places them in the hierarchy of norms and assesses their scope. The corpus comprises (i) the international

instruments binding on or relevant to Madagascar, notably the International Covenant on Economic, Social and Cultural Rights as interpreted by General Comment No. 15 [3], Agenda 21 [6], UN General Assembly Resolutions 64/292 and 70/1 [7], [8] and the Paris Agreement [9]; (ii) the Constitution of 11 December 2010 [10]; (iii) the Water Code [11] and its implementing decrees of 2003; (iv) the decentralization legislation, in particular Organic Law No. 2014-018 and Law No. 2014-020 [12], [13]; and (v) the texts that frame local and customary norms [14], [15]. Legal scholarship and sector reviews were used to document how these texts have been implemented [1], [2].

This doctrinal reading is combined with two other lenses. The first is comparative. South Africa, India, Indonesia and Kenya were selected because their courts or legislatures have taken explicit positions on the justiciability of the right to water and because they offer contrasting models: judicial review of reasonableness, judicial interpretation of the right to life, constitutional review of the opening of water to private exploitation, and explicit constitutional recognition combined with an independent regulator. The second lens is that of development economics, attentive to financing constraints, information asymmetries and the strategies of actors. The object itself calls for this combination: the effectiveness of a social right cannot be assessed by textual exegesis alone, and the analysis therefore moves from norms to institutions, and from institutions to practices.

2.2. Field Data: Sources, Collection and Validation

The empirical part relies on three secondary operational datasets on the Atsimo-Andrefana region. They were obtained from the Regional Directorate of Water, Sanitation and Hygiene (DREAH) of Atsimo-Andrefana during an interview with its Regional Director on 14 July 2026 [16]. Additional information was gathered from ADESO (*Association des Délégataires de l'Eau en Région Sud-Ouest*), the association of delegated water-service operators of the South-West region, from the DREAH, from JIRAMA and from the Ministry of Water, Sanitation and Hygiene (MEAH). (a) The monitoring file “Données ADESO AA” contains monthly readings for 2025 on twenty drinking-water supply sites: tariff per cubic metre, days of service, breakdowns and hours of unavailability, together with volumes produced and distributed and collection rates [17]. (b) The file “Situation des travaux Atsimo-Andrefana” records water-infrastructure worksites by district, technical partner and source of financing [18]. (c) The file “Donnée technique 165 MUS Atsimo Andrefana” describes 165 boreholes equipped as Multiple Use Services, with their depth, static level, exploitable flow and conductivity [19].

The authors did not draw a sample. The twenty sites are all the sites covered by the ADESO monitoring file for 2025, and the 268 worksites are all those recorded in the regional tracking file at the time of consultation. These are therefore exhaustive extractions of administrative records, not statistically representative samples of the region's water services. Because monitoring exists only where an operator and a reporting system are in place, the data probably give a more favourable picture of service quality than would be found in unmonitored areas.

The data were checked in three steps. Internal consistency rules were applied first: days of service could not exceed the number of days in the month, monthly unavailability could not exceed 744 hours, and tariffs had to fall within a plausible range; entries breaching these rules were treated as data-entry errors and left out of the calculations. The files were then cross-checked against one another (district names, partners, number of worksites per district). Finally, the results were compared with the qualitative findings of the 2019 national review of the Water Code's implementation [2]. Since several fields are incomplete (water produced, water distributed, collection rate), the analysis is limited to rounded descriptive statistics (ranges, averages and totals), and no inferential test was attempted. The incompleteness of the records is

itself treated as a finding, since it reveals the weakness of the sector's information system (Section 4.1.2).

2.3. Analytical Framework

Effectiveness is assessed against two grids. The normative grid derives from General Comment No. 15: availability, quality, physical and economic accessibility and non-discrimination, together with the State's obligations to respect, protect and fulfil [3, §§ 12, 20–29]. The institutional grid, drawn from institutional economics, examines the clarity of mandates, the match between competences and resources, and the information available to the public authority. A right is considered effective when its holders actually enjoy the guaranteed services and can obtain redress from an identifiable authority when the service fails.

3. THE LEGAL ENSHRINEMENT OF THE RIGHT TO WATER AND THE LIMITS OF MADAGASCAR'S INSTITUTIONAL FRAMEWORK

Malagasy water law is marked by a tension: a vigorous textual affirmation of a vital right (Section 3.1) is met by a fragile administrative architecture, characterized by overlapping competences and unfinished regulatory authorities (Section 3.2). The roots of ineffectiveness lie in the distance between the two.

3.1. The Textual Affirmation of a Vital Right

Three layers of norms combine to recognize the right to water in Madagascar: international law, which has shaped the whole construction; the Constitution; and the Water Code, which forms the core of the system. Scholarship has described the status given to water as a “bridging concept” linking the international and domestic legal orders [1, p. 67].

3.1.1.The International Anchoring of the Right to Water

International law has served as a matrix. At the 1992 Rio Conference, Chapter 18 of Agenda 21 set out a holistic vision of water, treated simultaneously as a social good, an economic good and an environmental resource, and promoted ecological management founded on the principles of environmental law, namely polluter-pays, precaution and prevention [6]. This matrix inspired the overhaul of national laws, including that of Madagascar.

A human right to water then took shape. Although the International Covenant on Economic, Social and Cultural Rights, to which Madagascar is a party, does not mention water, the Committee on Economic, Social and Cultural Rights derived such a right from it in General Comment No. 15 (2002). The Committee linked it to Article 11 (adequate standard of living) and Article 12 (health) and set out the criteria now used as benchmarks: availability, quality, physical and economic accessibility, and non-discrimination [3, §§ 2, 12]. In Resolution 64/292 of 28 July 2010, the UN General Assembly went further and recognized safe and clean drinking water and sanitation as a human right essential to the full enjoyment of life and of all human rights [7]. Sustainable Development Goal 6, which aims to ensure the availability and sustainable management of water and sanitation for all, now provides the programmatic framework; although it belongs to soft law, it informs the interpretation of States' obligations [8].

3.1.2.The Constitutional Foundation

In domestic law, the Constitution of 11 December 2010 does not expressly enshrine a right to water, but several of its provisions support one. Article 19 recognizes for every individual the right to the protection of health, of which access to safe water is a precondition. The Preamble commits the nation to preserving its natural wealth for future generations; Article 37 subjects freedom of enterprise to respect for the environment; Article 141 places the preservation of the environment among the competences that decentralized authorities exercise with the participation of the State; and Article 152 makes the *fokonolona*, the local community organized in *fokontany*, the basis of development and of environmental cohesion [10]. Above all, Article 137 gives duly published treaties an authority superior to that of statutes, so that the obligations arising from the Covenant, as interpreted by the Committee, rank above the Water Code itself. The right to water therefore rests on an indirect but supra-legislative foundation. Scholarship has shown that the notion of common heritage of the nation served as the vehicle through which the legislature brought the core principles of international development and environmental law into Malagasy law; its logic is teleological, since it places the person at the centre, and programmatic [1, pp. 66–68].

3.1.3.The Water Code as Pivotal Text

Law No. 98-029 of 20 January 1999 enacting the Water Code is the cornerstone of the system [11]. Its first article declares water to be part of the common heritage of the nation and makes the State its guarantor. Two major consequences follow. First, the resource is "communitarized": it is removed from absolute private appropriation and placed under a logic of national and intergenerational solidarity. Second, access to drinking water becomes a guiding principle of water management throughout the territory, which turns a statement of intent into a source of obligations borne by the State [1, p. 67].

The Code also organizes the threefold status of water that scholars have systematized: common heritage of the nation (collective dimension), attachment to the public domain (patrimonial and protective dimension), and a subjective right of access for each individual (personal dimension) [1, pp. 66–74]. It likewise adopts a modern approach to management: planning by hydrographic basins, special administrative police instruments (abstraction permits and declarations, classification of waters and regulation of discharges), and economic mechanisms (pricing, abstraction and discharge charges). On paper, therefore, Madagascar possessed as early as 1999 a water law regarded as exemplary in the sub-region.

Each component of this threefold status carries a distinct legal regime and distinct practical consequences. Communitarization grounds a form of solidarity but, in the absence of a clearly identified holder of rights, its legal scope remains uncertain, and scholars have long asked whether the formula is a mere declaration of principle or a legal notion in its own right [1]. Patrimonialization through the public domain offers more tangible protection: inalienability and imprescriptibility remove the resource from commerce and give the manager policing powers. It has a downside, however, since public-domain status imposes on the State a burden of management and upkeep that it cannot always assume, so that protection may turn into liability. Subjectivization, finally, has the most far-reaching consequences, because it makes the user the holder of enforceable rights; it is what gives the question of effectiveness its edge, since a subjective right without judicial guarantee remains virtual.

Justiciability is therefore decisive. A subjective right to water is meaningful only if its holder can obtain its respect before a court when the administration fails. In Madagascar, litigation on the right to water remains embryonic, for want of a culture of recourse and of effective access to the

administrative courts. The right to water remains largely a right without a judge, which leaves the administration a margin of inaction that no one sanctions. Comparative law shows that this situation is not inevitable (Section 3.1.4).

This formal quality makes the deficit of effectiveness all the more striking. The review carried out in 2019, twenty years after the Code's adoption, found that many of the structures meant to support its implementation had never been set up, that the hierarchy of texts had not always been respected and that roles overlapped [2, ch. 3, sects. 3.1–3.3]. The law exists; the administration meant to carry it is missing. Section 3.2 examines this finding, which is central to the argument.

3.1.4.Lessons from Comparative Law: Making the Right to Water Justiciable

Comparative experience shows that courts can give substance to the right to water, along different routes [20], [21]. In South Africa, Section 27(1)(b) of the 1996 Constitution guarantees everyone the right to have access to sufficient water, subject to progressive realization within available resources [22]. In *Mazibuko v. City of Johannesburg* (2009), the Constitutional Court declined to set a minimum quantity of water per person and upheld the city's free basic water policy and the installation of prepayment meters; it nevertheless affirmed that the State's measures are subject to judicial review of their reasonableness and must be kept under review [23]. The lesson lies less in the outcome than in the method: even a cautious court obliges the administration to justify its policy in public.

In India, where the Constitution contains no express right to water, the Supreme Court read one into the right to life protected by Article 21. In *Subhash Kumar v. State of Bihar* (1991), it held that the right to life includes the right to enjoy pollution-free water, and public interest litigation, with its broad rules on standing, has given this interpretation practical reach [24]. In Indonesia, the Constitutional Court, in Decision No. 85/PUU-XI/2013 of 18 February 2015, annulled the 2004 Water Resources Law in its entirety, holding that it did not sufficiently guarantee State control over water and the people's right to it in the face of private commercial exploitation [25], [21]. Kenya, finally, opted for explicit recognition: Article 43(1)(d) of the 2010 Constitution guarantees the right to clean and safe water in adequate quantities, and the Water Act of 2016 entrusts an independent regulatory board with overseeing service providers [26].

Three lessons can be drawn for Madagascar. First, an express constitutional or legislative right is not a prerequisite for justiciability: as the Indian route shows, a judge can derive it from existing guarantees, which in Madagascar could be the right to the protection of health (Article 19) read in the light of Article 137 and of General Comment No. 15. Second, review of reasonableness, as practised in South Africa, offers a realistic standard for a State with limited resources: it does not demand the impossible, but it does demand a plan, priorities and reasons. Third, the Indonesian and Kenyan examples show that justiciability also concerns the structure of the service, since courts or a regulator can check that contracting and pricing do not empty the right of its substance. These avenues presuppose effective access to the administrative courts and to constitutional review, which remains limited in Madagascar; the regulator discussed in Section 4.2.1 could offer users a first and more accessible forum for complaints.

3.2. The Shortcomings of the Administrative Architecture

The second source of fragility lies in the administrative architecture itself. The Water Code provided for an articulated institutional apparatus, resting on specialized authorities, economic regulation and a clear allocation of competences. In practice, this apparatus has remained partly virtual, or has generated paralysing overlaps.

3.2.1.Regulatory Authorities Provided For but Left Unfinished

The Water Code and its 2003 implementing decrees set up two key bodies. Decree No. 2003-192 of 4 March 2003 organized the National Water and Sanitation Authority (ANDEA), a public establishment designed to guarantee integrated resource management and planning by basin [27]. Decree No. 2003-939 of 9 September 2003 established SOREA, the regulator of the public water and sanitation service, and gave it responsibility for economic regulation: supervising tariffs, settling disputes between operators and users, and checking that service specifications are respected [28]. Decree No. 2003-191 of 4 March 2003 added basin agencies as the deconcentrated level of resource management [29].

Implementation did not match this design. Economic regulation remained largely theoretical because SOREA was never made operational on a lasting basis, and ANDEA never fully assumed the central role the Code assigned to it [2, sect. 3.1]. Tariffs, service quality and continuity are therefore subject to no independent and systematic control, which weakens users and operators alike. This is not a simple delay in execution but a recurrent feature of Malagasy administrative law. Creating a body by decree is cheap, quick and politically rewarding, whereas giving it qualified staff, a stable budget and real authority requires a long-term commitment that budget constraints and political instability discourage. The result is a series of “paper institutions”, legally existing but materially inert. Water law is a textbook case: the normative edifice is complete, while the administrative one is still a building site. A regulator that does not regulate is equivalent to no regulator at all, and the legal certainty it was meant to bring turns into insecurity.

Non-compliance with the hierarchy of texts aggravates this disorder. Because implementing measures were missing or inconsistent, some legislative provisions have remained unapplied, while administrative practices have developed alongside, or even against, the legal framework, which raises a problem of legitimacy for public interventions [2, sect. 3.2]. This breach of the principle of legality, cardinal in administrative law, has consequences for the right to water: it deprives users of a stable and predictable framework and exposes public and private investment alike to a legal risk that raises its cost and discourages it.

3.2.2.The Overlapping of Ministerial Competences

The fragility of the specialized authorities is compounded by the instability of the sector's ministerial anchoring. From one government reshuffle to the next, the water portfolio has been attached to different departments (energy, water, sanitation), sometimes split among several ministries, without clear boundaries of competence. This instability has two harmful effects. First, roles and responsibilities overlap among administrations, which produces duplication, conflicts of competence and diluted responsibility [2, sect. 3.3]. Second, public policies become discontinuous: each reorganization interrupts ongoing work and raises coordination costs. In the terms of institutional economics, the architecture lacks clear mandates, which are the first condition of the accountability of actors.

3.2.3.The Ambiguous Position of JIRAMA

The national utility JIRAMA occupies a singular and ambiguous place in this landscape. As the historic operator of water and electricity in urban areas, it ensures production and distribution in the main towns but accumulates difficulties: chronic under-investment, high technical and commercial losses, tariffs disconnected from costs, and dependence on public subsidies. Its situation illustrates the confusion of roles characteristic of the sector: it is at once operator, de facto quasi-regulator in its service areas, and instrument of the government's tariff policy. This

confusion runs counter to the separation of regulation and operation laid down by the Code. It also deprives users of an effective remedy: without an operational regulator, the relationship between subscriber and operator remains deeply unequal.

The financial fragility of the historic operator is not a mere management issue: it directly affects the effectiveness of the right. A structurally loss-making operator can neither maintain its network properly nor extend it to unserved neighbourhoods, which are most often the poorest. Under-investment translates into heavy network losses, unbilled water that weighs on the economic balance, and de facto rationing that hits the most vulnerable users. The circle is vicious: weak tariffs compromise investment, lack of investment degrades the service, a degraded service undermines willingness to pay, and falling revenue further weakens tariffs. Breaking this circle requires coherent public intervention combining recapitalization, tariff reform and regulation.
Section 3 thus leads to a clear finding: Malagasy water law is not deficient in its letter, and its weakness lies in the incompleteness of the administration meant to implement it. Section 4 examines how this weakness plays out on the ground.

4. THE CHALLENGES OF PUBLIC ACTION: FROM NATURAL-RESOURCE MANAGEMENT TO THE EFFECTIVENESS OF THE RIGHT

Section 3 identified an institutional incompleteness at the top; this section examines its effects at the base. It considers the inability of decentralized authorities to run the public water service transferred to them (Section 4.1), the interface between statutory and customary water governance (Section 4.1.4), and the avenues for redesigning governance so as to make the right effective (Sections 4.2 and 4.3). The analysis draws on the Atsimo-Andrefana data described in Section 2.2.

4.1. The Paralysis of Decentralization

Making municipalities the contracting authorities for water services was one of the main orientations of sector policy. Inspired by subsidiarity, it assumes that the level closest to users is best placed to identify needs and manage the service. The transfer of competence, however, was not matched by a transfer of resources.

4.1.1.A Competence Transferred Without the Corresponding Means

Municipal contracting authority over drinking-water supply derives from the Water Code and from its decree on the public water and sanitation service [11], [30]. The decentralization legislation confirmed it: first Law No. 94-007 of 26 April 1995, then Organic Law No. 2014-018, whose Article 28 lists water and sanitation among the operations falling within municipal development, and Law No. 2014-020 on the resources of decentralized authorities [31], [12], [13]. The financial, human and technical resources needed to exercise this competence did not follow. The principle of concomitance, under which every transfer of competence must come with a corresponding transfer of resources, has remained a dead letter: the average rural municipality has neither the engineering skills, nor the tax base, nor the procurement capacity that contracting authority over a supply scheme requires.

The result is a contracting authority “in trompe-l'œil”: the municipality holds the competence in law but cannot exercise it in fact. Public-finance theory calls this an “unfunded mandate”, a legal obligation without a budgetary counterpart, which mechanically leads to under-provision of the service. With negligible own revenue and no borrowing capacity, the rural municipality can neither invest, nor operate the works, nor recruit the skills a water service requires

(hydrogeology, civil engineering, accounting and commercial management, contract design). The chain of responsibility breaks as a consequence: users deprived of water no longer know whom to hold accountable, since the State points to municipal competence and the municipality to its lack of means. Decentralization, meant to bring decisions closer to needs, has turned into a transfer of powerlessness.

4.1.2. Information Asymmetry and the Deficit of Economic Steering

The insufficiency of means is compounded by a structural information asymmetry that hampers the economic steering of the service. The local authority, like the supervising administration, has only a very partial grasp of the basic data (volumes produced and distributed, real operating costs, collection rate, state of the assets) without which neither rational tariff regulation nor investment planning is possible. The operating data collected in Atsimo-Andrefana confirm this: records are incomplete, heterogeneous from one site to another and at times marred by obvious data-entry errors, a sign of a still embryonic information system [17]. In development economics, information is the first condition of decision: without reliable data, the public authority acts blindly and is disarmed before the operator, who alone holds detailed knowledge of operations.

Despite their limitations, the available data reveal characteristic tensions. Table 1 summarizes the main operating indicators for the twenty drinking-water supply sites monitored in 2025.

Table 1. Operating indicators for twenty monitored drinking-water supply sites (Atsimo-Andrefana, 2025)

Indicator	Observed value	Interpretation
Number of sites monitored	20	Partial coverage of the regional territory
Water price at connection (Ar/m³)	1,500 to 5,000	Wide tariff dispersion, unregulated
Continuity of service (days/month)	29 on average	Good in appearance, but…
Breakdowns recorded (monthly total)	> 400	…high frequency of incidents
Maximum unavailability (hours/month)	up to 744	Equivalent to a full month of stoppage at a site

Source: compiled by the authors from the monitoring file "Données ADESO AA" (2025) [17]; see Section 2.2 for the validation procedure. Values rounded; some records are incomplete.

Two lessons emerge. First, the tariff dispersion, from 1,500 to 5,000 ariary per cubic metre depending on the site, reflects the absence of economic regulation: each operator sets its price according to its costs and margin, without a reference schedule or control by an independent regulator. This heterogeneity raises a question of territorial equity: the right of access, formally universal, depends on users' ability to pay and on where they live, contrary to the criterion of economic accessibility laid down by international law [3, § 12(c)(ii)] and to the equity concerns raised in the national review [2, sect. 3.5]. Second, the reliability of the service is far more fragile than the displayed continuity suggests: the frequency of breakdowns and periods of unavailability that can reach the equivalent of a full month at certain sites reveal a deficit of maintenance and sustainability, a recurrent weakness of rural water services in Madagascar [2, sect. 3.6].

4.1.3. The Substitution of Donors and NGOs for Public Authority

From the standpoint of public-service theory, the most significant effect is that technical and financial partners and non-governmental organizations have, in practice, taken the place of the

public authority. The status of water-infrastructure works in Atsimo-Andrefana is telling in this respect: nearly all recorded worksites are carried by non-State actors.

Table 2. Distribution of water-infrastructure worksites by technical partner (Atsimo-Andrefana)

Technical partner	**Number of worksites**	**Share (%)**
UNICEF	169	63
Action Against Hunger (ACF)	67	25
Red Cross	23	9
Expert Solidaire	6	2
TRANSMAD	3	1
Total recorded	268	100

Source: compiled by the authors from the file "Situation des travaux Atsimo-Andrefana" [18]. Financing mainly provided by UNICEF, USAID and the Mionjo programme.

The 268 worksites are distributed over seven districts and concentrated in the most vulnerable ones: Betioky accounts for 68, Ampanihy and Morombe for 32 each, Ankazoabo for 21 and Toliara II for 20, whereas Beroroha (5) and Benenitra (4) are barely covered [18]. Funding comes mainly from UNICEF, USAID and the Mionjo programme, and the works are implemented by agencies and NGOs, together with operators such as Young Progress; a single project template foresees a 2026 budget of about USD 195,000 for one "WASH Emergency" component [18]. In addition, 165 boreholes have been equipped as Multiple Use Services (MUS); their technical records show the scale of the effort as well as its reliance on external funding [19].

This finding calls for a rigorous legal reading. The public water service, a general-interest activity par excellence, is in fact provided by private and international actors, with external funding, and often within a humanitarian-emergency framework (the nomenclatures speak of "WASH Emergency") rather than as a durable public service. Humanitarian intervention, which is by nature occasional and depends on the availability of funding, cannot stand in for public policy. It also raises the question of sustainability: what becomes of the infrastructure once donors leave, if no public authority is able to operate, maintain and renew it? Substitution relieves the emergency, but it perpetuates the structural deficiency of the public authority and postpones the construction of a sustainable public service. It also reveals a form of withdrawal by the State from its role as guarantor of the vital right to water.

This configuration also sits uneasily with the founding principles of public service. The so-called Rolland laws (continuity, equality and adaptability) presuppose an identifiable organizing authority that guarantees the permanence of the service and the equal treatment of users. When the service depends on the renewal of external funding and on operators whose mandate is by definition temporary, continuity is no longer legally guaranteed but merely hoped for. Likewise, the geography of intervention, dense where donors choose to act and nearly absent elsewhere, introduces a territorial inequality that no public authority corrects. The right to water then depends on decisions taken outside the national public authority, which in the long run raises a question of sovereignty over a vital sector. The generosity of partners does not exempt the State from rebuilding a public service of which it remains, in law, the ultimate debtor.

4.1.4.Customary Water Governance and Legal Pluralism

The rural focus of the study requires attention to a dimension that formal law tends to overlook. In the South-West, access to water points is governed not only by the Water Code but also by

local norms: collective decisions of the *fokonolona*, rules of precedence that favour lineages regarded as *tompon-tany* (“masters of the land”, the first occupants), prohibitions (*fady*) attached to certain springs, lakes or wells, and conventions (*dina*) that set rules of use and sanctions. These arrangements determine in practice who may draw water, at what times, for which uses (household needs, livestock watering, small-scale irrigation) and who must contribute to maintenance. The Multiple Use Services recorded in the region [19] answer precisely this plurality of uses, which the sectoral logic of the Code, separating drinking water from agricultural water, captures poorly.

Legal anthropology describes this situation as legal pluralism: several normative orders coexist, overlap and are invoked selectively by actors according to their interests [32], [33]. Malagasy law already contains instruments for articulating them. Law No. 96-025 of 30 September 1996 (GELOSE) allows the management of renewable natural resources to be transferred by contract to local community associations [14]. Law No. 2001-004 of 25 October 2001 regulates *dina*, which acquire binding force once homologated by the court, provided they comply with the Constitution and statutes [15]. The Constitution itself, in Article 152, makes the *fokonolona* the basis of development and of environmental cohesion [10].

Three proposals follow from this. First, water-point management committees could be recognized as the community tier of the public service, through a contract with the municipality, as contracting authority, that sets out their tasks, the rules on tariffs or contributions, and the technical support they receive. Second, local rules of use could be formalized as *dina* homologated by the court; this would give them legal force while subjecting them to review of their conformity with the Constitution, in particular the principle of non-discrimination, since customary hierarchies of access may disadvantage women, migrants or lineages of lower status, and the right to water cannot yield to them. Third, basin and municipal planning should include an inventory of customary water points and of the rules attached to them. On these conditions, water law would no longer be imposed from above but would be appropriated by those it governs.

4.2. Toward a Redesign of Water Governance: Legal and Political-Economy Avenues

With the diagnosis established, the avenues for an effective guarantee can be sketched. They do not lie in further normative inflation, since the existing corpus is already substantial, but in a redesign of governance that combines institutional clarification, secure and climate-informed financing, controlled contracting and a stronger water police, supported by a cross-cutting requirement of information and participation.

4.2.1.Clarifying the Architecture and Operationalizing Regulation

The first requirement is institutional. The task is less to create new bodies than to make effective those the law already provides for [2, ch. 4]. The priority is to operationalize an independent economic regulator with real resources and legally guaranteed powers, sheltered from the hazards of reshuffles. Such a regulator would perform three functions: supervising tariffs, so as to correct the dispersion and inequity noted above; controlling the quality and continuity of the service; and arbitrating disputes between users and operators. Its indispensable corollary is a text that stabilizes the sector's ministerial anchoring and fixes competences durably.

The regulator's independence is not a formality. It conditions the credibility of regulation, because a body subject to political power or captured by the operator cannot arbitrate impartially between the interest of users, the sustainability of the operator and the constraints of the State

budget. Regulation theory calls for concrete guarantees: fixed-term mandates for directors, not revocable at will; dedicated own resources; effective powers of investigation and sanction; and an obligation to give reasons for decisions and to publish them. The function of economic regulation (control of the service) should also be clearly distinguished from that of resource management (water police and basin planning), so that the confusion of roles criticized at the base is not reproduced at the top.

In concrete terms, operationalizing SOREA would involve: (i) revising Decree No. 2003-939 to secure the regulator's independence and its powers of investigation and sanction; (ii) appointing a board for a fixed, non-renewable term, whose members cannot be dismissed except for serious misconduct; (iii) providing own resources, for instance through a regulatory levy set at a small percentage of operators' turnover, supplemented by a State allocation during the start-up phase; (iv) recruiting a small core team combining regulatory economics, engineering and law; and (v) publishing a reference tariff schedule and annual performance reports for the sites under its jurisdiction. Table 3 (Section 4.3) sets out the sequence and the indicators.

4.2.2.Securing Sustainable Financing of the Resource and the Service

The second requirement is financial. The effectiveness of a social right has a cost, and the sustainability of the service presupposes a balanced economic model, whereas sector financing has long been insufficient [2, sect. 3.4]. Three levers deserve to be used. The first is the effective collection of the economic instruments already provided for by the Code, namely abstraction and discharge charges and incentive pricing, with their proceeds allocated to the sector [34], [35]. The second is the creation and funding of a national fund dedicated to water resources, an equalization instrument between surplus and deficit territories capable of correcting the spatial inequalities revealed by the case study. The third is a medium-term expenditure framework specific to the sector, which would give investors predictability and policy continuity.

Pricing calls for particular attention, because it concentrates the tension between economic sustainability and social accessibility. The principle of cost recovery, indispensable to durability, must be reconciled with the economic accessibility required by the fundamental right. Social-tariff arrangements, such as a first block of vital consumption at a moderate price or subsidies targeted at the most vulnerable, reconcile the two without sacrificing either [3, § 27]. The sector's current dependence on external aid, although it brings valuable resources, is not a sustainable long-term financing model: it exposes water policy to shifts in donors' priorities and deprives the State of control over its own strategy. Building domestic financing, however gradually, is the condition of genuine national ownership of the right to water.

4.2.3.Mastering Contractualization and Public-Private Partnership

The third avenue, the most delicate, is contractualization. Given the incapacity of local authorities and the fragility of the public operator, recourse to the private sector through delegation of public service, lease contracts (*affermage*) or public-private partnership appears a pragmatic path, already tried in Madagascar [2, sect. 2.6]. These arrangements bring technical capacity and private capital and introduce a logic of performance. They nonetheless carry risks well identified in the scholarship of public economic law: capture of rents by the operator, information asymmetry to the detriment of the public contracting party, and exclusion of users unable to pay.

Contractualization cannot therefore be an abdication of the public authority; it is, on the contrary, a demanding way of exercising it. Its success depends on three conditions: a competent public contracting party, able to negotiate and monitor the contract on equal terms with the operator, which requires stronger municipal capacities and State support; an effective regulator, guarantor

of the balance of the contract and of users' rights; and robust public-service clauses covering coverage obligations, regulated tariffs, quality and continuity standards, and sanctions for default. Otherwise, partnership may turn a failing public monopoly into an unregulated private one, at the expense of the very right it is meant to guarantee.

Scattered rural settings such as Atsimo-Andrefana also call for adapted arrangements, because low density and modest incomes undermine the profitability of models designed for urban areas. Intermediate formulas deserve to be explored: grouping several water points in a single contract to pool fixed costs and reach a critical size; community management that is supervised, professionalized and backed by durable technical support; or delegation to a regional operator covering several municipalities. Whatever the formula, the public authority must keep control over the definition of public-service obligations and have the means to monitor their execution. Contractualization changes the mode of management; it never transfers ultimate responsibility for the right, which remains with the public guarantor.

4.2.4.Strengthening the Water Police and the Protection of the Resource

The fourth avenue, often neglected, concerns the protection of the resource itself, upstream of the service. The water police, a special administrative-police power that regulates abstractions, protects catchment points and represses pollution, remains weakly exercised for want of means of control and sanction. Yet in a context of climate-driven scarcity, the quantitative and qualitative protection of the resource becomes a matter of security. The instruments exist: classification of surface waters and regulation of liquid-effluent discharges [36], rules on discharges and deposits into surface water and groundwater [37], the abstraction-permit procedure [38] and protection perimeters around catchments [39]. They need to be applied, which requires inspectors, registers and sanctions. The water police is the sovereign side of the right to water: without a protected resource, there is no durable service.

4.2.5.Integrating Climate Adaptation and Mobilizing Climate Finance

Climate change gives these orientations particular urgency. In the South-West, more frequent droughts and a disrupted rainfall calendar increase the vulnerability of agro-pastoral systems [4] and sharpen competition between uses. Water law is thus called upon to arbitrate and to prevent conflicts: by giving priority to human consumption, by granting enhanced protection to strategic resources, and by integrating climate variability into basin planning. Resource management then becomes a matter of justice between territories, between uses and between generations.

The legal framework has begun to take this into account. A review carried out in 2020 found that the Code's general provisions on integrated water resources management lacked precision on climate change; it proposed an implementing decree on an adaptation strategy for the water, sanitation and hygiene sector, standards for climate-resilient infrastructure, the integration of climate impacts into the sector database, and early-warning systems at basin-agency level [40]. Madagascar's third Nationally Determined Contribution, covering 2025–2035, sets the national framework into which these measures should fit [41]. Three legal reforms would help the water sector benefit from it.

First, international climate finance, organized under Articles 7 and 9 of the Paris Agreement [9], is channelled through funds such as the Green Climate Fund and the Adaptation Fund, which require projects to be submitted through accredited entities that meet fiduciary, environmental and social safeguards. Accrediting a national entity for direct access, and aligning public procurement and financial-management rules with these standards, would reduce dependence on international intermediaries. Second, climate resources mobilized for water should flow through

the national water fund proposed above (Section 4.2.2), under rules that ring-fence them for investment and maintenance in the most exposed areas, rather than through isolated projects. Third, the Water Code should provide for a drought regime: a legal basis allowing the administration to declare water stress by basin, to restrict non-priority abstractions and to guarantee a minimum supply for domestic use and livestock watering, under the control of the courts.

4.2.6. A Cross-Cutting Requirement: Information, Accountability and Participation

None of these orientations will bear fruit without a foundation often held to be secondary: the production and sharing of reliable information. As shown above, information asymmetry disarms the local authority before the operator and rules out serious regulation. A national water-information system, comprising an inventory of works, monitoring of volumes and quality, analytical accounts of services and a map of service coverage, is therefore not an administrative convenience but part of the infrastructure of governance. It conditions the accountability of actors, the objectivity of budget decisions and the measurement of progress toward universal access. The field data examined here, valuable but incomplete and heterogeneous, show by contrast what the absence of such a system costs: public action proceeds without a dashboard, and its effectiveness cannot be evaluated.

User participation is the second pillar of this requirement. The right to water cannot be reduced to a vertical relationship between administration and users; it presupposes the involvement of communities in setting priorities, overseeing management and maintaining the works. Community-management arrangements, when supervised and supported, improve the durability of installations and strengthen willingness to pay. At the local scale, they express the collective dimension of the common heritage of the nation: since the resource belongs to all, its governance cannot be confiscated. This participatory approach is consistent with human-rights-based guidance, which treats participation, transparency and accountability as components of the right to water [42].

4.3. From Principles to Action: An Operational Roadmap

Table 3 translates these orientations into concrete steps and specifies, for each of them, the lead and partner actors, an indicative timeline and indicators of success. The timelines are indicative: they assume a political commitment that the study cannot take for granted, and they should be refined in consultation with the sector's stakeholders.

Table 3. Operational roadmap for redesigning water governance

Orientation	Concrete steps	Lead and partner actors	Indicative timeline	Indicators of success
1. Operational regulator	Revise Decree No. 2003-939; appoint a board with fixed, non-renewable terms; create a regulatory levy and a start-up State allocation; recruit a core team (economists, engineers, lawyers); publish a reference tariff schedule	Ministry in charge of water (lead); Ministry of Finance; SOREA; operators	0–2 years (decree, board, team); 2–4 years (tariff schedule)	Board appointed and budget executed; share of sites applying the reference schedule; number of user complaints handled per year

Orientation	Concrete steps	Lead and partner actors	Indicative timeline	Indicators of success
2. Sustainable financing	Collect abstraction and discharge charges; capitalize a national water fund with equalization rules; adopt a sector medium-term expenditure framework; introduce a vital consumption block in tariffs	Ministry in charge of water (lead); Ministry of Finance; ANDEA; municipalities	1–3 years	Share of charges actually collected; share of domestic resources in sector investment; price of the first consumption block relative to household income
3. Municipal contracting	Group water points into inter-municipal contracts; set up regional technical-support units; adopt standard delegation contracts with public-service clauses; train municipal staff	Municipalities (contracting authorities); regional directorates; ministry in charge of decentralization	1–3 years	Number of contracts signed on the standard model; share of sites with an operator under contract; average duration of breakdowns
4. Water police	Keep a register of abstraction permits; delimit protection perimeters around catchments; deploy sworn inspectors at basin level; apply graduated sanctions	ANDEA (lead); basin agencies; regional directorates	2–5 years	Number of permits registered; share of catchments with a protection perimeter; number of inspections and sanctions per year
5. Information and participation	Build a national water-information system (works, volumes, quality, accounts); publish an annual report per site; give user committees a role in monitoring	Ministry in charge of water (lead); SOREA; operators; partners and NGOs	0–3 years	Share of sites reporting complete monthly data; annual reports published; user committees in operation
6. Customary interface	Inventory customary water points and rules; formalize rules of use as homologated dina; sign contracts between municipalities and management committees; check non-discrimination	Municipalities (lead); fokonolona; courts (homologation); regional directorates	1–3 years	Number of homologated dina on water; share of committees with women members; access disputes resolved
7. Climate adaptation and finance	Adopt the decree on the sector adaptation strategy; seek accreditation of a national entity for direct access; open a climate window in the water fund; create a legal drought regime	Ministry in charge of water (lead); Ministry of Environment; Ministry of Finance	1–5 years	Decree adopted; accreditation obtained; climate finance channelled to water; minimum supply ensured during declared droughts

Source: authors' proposals based on Sections 3 and 4. Timelines are indicative and run from the adoption of the reform.

5. LIMITATIONS OF THE STUDY

The findings should be read in the light of several limitations. First, the field data cover a single region, Atsimo-Andrefana, whose semi-arid conditions and scattered settlement make it a revealing case but not a representative one: urban services operated by JIRAMA, or the more humid regions of the East and the Highlands, may present different configurations. Second, the data are secondary operational records produced for monitoring rather than for research; they cover only the sites where a monitoring system exists, relate essentially to 2025 and contain incomplete fields, which limits the analysis to descriptive statistics and precludes causal inference.

Third, part of the institutional diagnosis relies on secondary sources, in particular the 2019 review of the Water Code's implementation [2], whose findings the authors could not verify independently for everybody and every text; the situation may have changed since, notably with the ongoing work on revising the Code [40]. Fourth, the comparative analysis is deliberately selective and does not claim that foreign solutions can be transposed without adaptation. Finally, the recommendations reflect the authors' legal and economic analysis; apart from the interview with the DREAH Regional Director, they have not been validated through a structured consultation of the sector's stakeholders, whose views may differ. Future research could extend the analysis to other regions, combine administrative data with household surveys and interviews with local officials, and examine the case law of Malagasy courts on water disputes.

6. CONCLUSION

At the close of this analysis, the Malagasy paradox of the right to water can be clearly grasped. Madagascar has an ambitious water law, supported by the Constitution, structured by a Code regarded as modern and aligned with international human rights standards. The resource is legally qualified as the common heritage of the nation, and access to drinking water is treated as a fundamental right. On the terrain of norms, the edifice is solid; on that of effectiveness, it wavers. The analysis has shown that this ineffectiveness stems from a deficit of administration rather than of legislation. The Atsimo-Andrefana case gives this diagnosis an empirical basis: a service delivered mostly by donors and NGOs, with external funding and in an emergency logic; dispersed and inequitable tariffs; reliability undermined by poor maintenance; and records too incomplete to steer public action. The public authority, guarantor in law of the vital right to water, is largely absent in fact.

An effective guarantee therefore requires reactivating existing instruments rather than adopting new texts: an operational regulator, sustainable and climate-informed financing, controlled contracting, a stronger water police, and recognition of customary water institutions within a framework that respects non-discrimination. Comparative law shows that courts and regulators can make the right justiciable even where resources are limited, provided the administration is required to plan and to justify its choices. These reforms belong as much to administrative law as to development economics, and they require a clear political choice: making water a genuine priority of public action.

Beyond the water sector, the study invites broader reflection on the effectiveness of social rights in developing States: a right exists fully only through the institutions that carry it, and a proclamation is worth no more than the administration in charge of implementing it. As climate pressure makes water scarcer and conflicts between uses more acute, building water governance that is fair, sustainable and effective has become a condition of development that respects human

dignity. In Madagascar, the vital right to water awaits less new promises than a public authority able to keep those already made.

GLOSSARY OF MALAGASY TERMS AND ACRONYMS

Term or acronym	Meaning
AA	Atsimo-Andrefana (abbreviation used in the regional monitoring files)
ACF	Action Against Hunger (Action contre la Faim), international NGO
ADESO	Association des Délégataires de l'Eau en Région Sud-Ouest (association of delegated water-service operators of the South-Westregion)
DREAH	Direction Régionale de l'Eau, de l'Assainissement et de l'Hygiène (RegionalDirectorate of Water, Sanitation and Hygiene), Atsimo-Andrefana
ANDEA	Autorité Nationale de l'Eau et de l'Assainissement (National Water and SanitationAuthority)
Ariary (Ar)	National currency of Madagascar
Atsimo-Andrefana	Region of South-West Madagascar (atsimo: south; andrefana: west), capital Toliara
CDN / NDC	Contribution Déterminée au niveau National (Nationally Determined Contribution) under the Paris Agreement
CESCR	UN Committee on Economic, Social and Cultural Rights
CTD	Collectivité Territoriale Décentralisée (decentralized territorial authority: region, municipality)
Dina	Local convention adopted by a community, setting rules and sanctions; binding once homologated by the court
Fady	Customary prohibition or taboo, which may attach to a place, a practice or a resource such as a spring
Fokonolona	Local community of a village or neighbourhood; recognized by the Constitution as the basis of development
Fokontany	Smallest administrative subdivision, within a municipality
GELOSE	Gestion Locale Sécurisée: contractual transfer of the management of renewable natural resources to local communities (Law No. 96-025)
JIRAMA	Jiro sy Rano Malagasy, literally “Malagasy Electricity and Water” (jiro: light, electricity; rano: water), the national utility
MEAH	Ministère de l'Eau, de l'Assainissement et de l'Hygiène (Ministry of Water, Sanitation and Hygiene)
MUS	Multiple Use Services: water points designed for several uses (domestic, livestock, small-scale irrigation)
NGO	Non-governmental organization
SDG	Sustainable Development Goal (SDG 6: clean water and sanitation)

Term or acronym	Meaning
SOREA	Organisme Régulateur du Service Public de l'Eau et de l'Assainissement (Regulatory Body for the Public Water and Sanitation Service)
Tompon-tany	"Masters of the land": lineages regarded as first occupants, who often hold precedence in access to resources
WASH	Water, Sanitation and Hygiene

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